# Kishor Jothimurugan

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## Education

**University of Pennsylvania** **Philadelphia, USA**
*Ph.D. in Computer and Information Science, GPA 4/4* *2017–2023*
Thesis Topic: Specification-Guided Reinforcement Learning
Committee: Dr. Osbert Bastani, Dr. Thomas A. Henzinger, Dr. Sampath Kannan, and Dr. George J. Pappas
**Advised by Dr. Rajeev Alur**

**Chennai Mathematical Institute** **Chennai, India**
*B.Sc. (Honors) in Mathematics and Computer Science, CGPA 9.77/10* *2014–2017*
**Ranked among top 3 students**

## Work Experience

**Two Sigma Investments, LP** 2023–Current
*Quantitative Researcher*
Systematic macro research

**Amazon Web Services** Summer 2022
*Applied Scientist Intern, AI Labs*
Incorporating program semantics in transformer-based code generation models
Mentors: Dr. Nathan Fulton, Dr. Siddhartha Jain and Dr. Baishakhi Ray

**Nokia Bell Labs** Summer 2020
*Research Intern, Data and AI Lab*
An application of deep RL to regenerative stopping problems
Mentors: Dr. Matthew Andrews, Dr. Jeongran Lee and Dr. Lorenzo Maggi

**Amazon Web Services** Summer 2019
*Software Development Engineering Intern, Automated Reasoning Group*
Using machine learning to improve usability of taint analysis
Mentors: Dr. Andrew Gacek and Dr. Lee Pike

**ENS Paris-Saclay** Summer 2017
*Research Intern, Formal Methods Group*
Mentor: Dr. Dietmar Berwanger

## Publications

First (co)author in all publications; names are listed alphabetically in some papers
* equal contribution, † authors in alphabetical order

## Journal Publications

**[CACM 26]** Specification-Guided Reinforcement Learning
R. Alur, S. Bansal, O. Bastani and *K. Jothimurugan*†
*Communications of the ACM, 2026*

## Conference Publications

**[NeuS 25]** Specification-Guided Reinforcement Learning
*K. Jothimurugan*, S. Bansal, O. Bastani and R. Alur
*International Conference on Neuro-symbolic Systems, 2025*

**[ICML 23]** Robust Subtask Learning for Compositional Generalization
*K. Jothimurugan*, S. Hsu, O. Bastani and R. Alur
*International Conference on Machine Learning, 2023*
**Acceptance rate: 27.9%**

**[CAV 23]** Policy Synthesis and Reinforcement Learning for Discounted LTL
R. Alur, O. Bastani, *K. Jothimurugan*†, M. Perez, F. Somenzi and A. Trivedi
*International Conference on Computer Aided Verification, 2023*
**Acceptance rate: 26%**

**[Festschrift 22]** A Framework for Transforming Specifications in Reinforcement Learning
R. Alur, S. Bansal, O. Bastani and *K. Jothimurugan*†
*Springer Festschrift in honor of Prof. Tom Henzinger (Invited), 2022*

**[CAV 22]** Specification-Guided Learning of Nash Equilibria with High Social Welfare
*K. Jothimurugan*, S. Bansal, O. Bastani and R. Alur
*International Conference on Computer Aided Verification, 2022*
**Acceptance rate: 24.4%**

**[NeurIPS 21]** Compositional Reinforcement Learning from Logical Specifications
*K. Jothimurugan*, S. Bansal, O. Bastani and R. Alur
*Advances in Neural Information Processing Systems, 2021*
**Acceptance rate: 25.7%**

**[EMSOFT 21]** Compositional Learning and Verification of Neural Network Controllers
R. Ivanov*, *K. Jothimurugan**, S. Hsu, S. Vaidya, R. Alur and O. Bastani
*International Conference on Embedded Software, 2021*
**Acceptance rate: 25.2%**

**[AISTATS 21]** Abstract Value Iteration for Hierarchical Reinforcement Learning
*K. Jothimurugan*, O. Bastani and R. Alur
*International Conference on Artificial Intelligence and Statistics, 2021*
**Acceptance rate: 29.8%**

**[LICS 20]** Space-efficient Query Evaluation over Probabilistic Event Streams
R. Alur, Y. Chen, *K. Jothimurugan*† and S. Khanna
*Symposium on Logic in Computer Science, 2020*
**Acceptance rate: 39.6%**

**[NeurIPS 19]** A Composable Specification Language for Reinforcement Learning Tasks

*K. Jothimurugan*, R. Alur and O. Bastani
*Advances in Neural Information Processing Systems, 2019*
**Acceptance rate: 21.1%**

### Preprints and Technical Reports

Learning Algorithms for Regenerative Stopping Problems with Logistics Applications
*K. Jothimurugan*, M. Andrews, J. Lee and L. Maggi
*Nokia Bell Labs Intern Report*

## Tutorials

**[AAAI 23]** Specification-Guided Reinforcement Learning
Presenters: *K. Jothimurugan*, S. Bansal, R. Alur and O. Bastani
*AAAI Conference on Artificial Intelligence, 2023*
A comprehensive tutorial on RL algorithms for learning control policies from logical specifications

## Open Source Tools

**DIRL** — Github link
Compositional RL algorithm for learning from temporal specifications

**HIGH-NASH** — Github link (CAV Artifact Evaluation Badge: Functional)
Multi-agent RL algorithm for learning Nash equilibria with high social welfare

**SARL** — Github link
Hierarchical RL algorithm leveraging user provided state abstractions

**SPECTRL** — Github link
Generating shaped rewards from temporal specifications

## Teaching Experience

### Guest Lecturer

Computer-Aided Verification, CIS 673 (Class of $\sim 20$) *Fall 2021*
Graduate Course, University of Pennsylvania
Topic: Techniques for Verifying Robustness of Neural Networks

### Teaching Assistant

Principles of Embedded Systems, CIS 540 (Class of $\sim 20$) *Spring 2019*
Graduate Course, University of Pennsylvania

Automata, Computability and Complexity, CIS 262 (Class of $\sim 150$) *Fall 2018*
Undergraduate Course, University of Pennsylvania

Discrete Mathematics (Class of $\sim 30$) *Spring 2017*
Undergraduate Course, Chennai Mathematical Institute

Design and Analysis of Algorithms *Fall 2016*
Online Course offered by Indian Institute of Technology, Madras

## Mentoring and Outreach

**Graduate Student Mentoring** 2020–2022
*Steve Hsu, Masters Student, University of Pennsylvania*
Topic: Compositional reinforcement learning for multi-task generalization
**Co-authored two papers**

**Penn CIS Mentorship Program** 2022
*Participated as a mentor*
Department-wide effort to connect early career PhD students with senior PhD students
Attracted participation from 34 first-year students

## Awards

**CTL Teaching Certificate** *Spring 2022*
Awarded by Center for Teaching and Learning, University of Pennsylvania

**CMI Undergraduate Scholarship** *2014–2017*
Awarded by CMI to undergraduate students for excellence in academics

## Research Presentations

### Invited Talks

**IARCS Verification Seminar Series** *Summer 2023*
Title: Specification-Guided Reinforcement Learning

**University of Illinois Urbana-Champaign** *Spring 2023*
Title: Specification-Guided Reinforcement Learning

**University of Virginia** *Spring 2023*
Title: Specification-Guided Reinforcement Learning

**NYU Formal Methods Group** *Fall 2022*
Title: Specification-Guided Reinforcement Learning

**Microsoft PROSE Group** *Fall 2022*
Title: Specification-Guided Reinforcement Learning

**Thomas Henzinger's Group, IST Austria** *Fall 2021*
Title: Reinforcement Learning from Logical Specifications

**Simons Institute, UC Berkeley** *Spring 2021*
Workshop on Games and Equilibria in System Design and Analysis
Title: Abstract Value Iteration for Hierarchical Reinforcement Learning

### Presentations and Posters (from publications)

**Conferences:** NeurIPS 2019 and 2021, LICS 2020, AISTATS 2021, EMSOFT 2021, CAV 2022, ICML 2023.

**Workshops:** DeepRL @ NeurIPS 2020 and 2022, SafeRL @ NeurIPS 2021, SYNT @ CAV 2022.

## Review Service

**Conferences:** NeurIPS 2022, AAAI 2023, AISTATS 2023, ESOP 2023, L4DC 2023, ICML 2023, CAV 2023, NeurIPS 2023, ICLR 2024, ICML 2024, NeurIPS 2024, AAAI 2025, ICML 2025.

**Journals:** IEEE TCAD, JAIR.